[Total No. of Questions - 9] [Total No. of Printed Pages - 4]

Dec-24-0354 (NEP)

MCA-7306(v) (Software Engineering)

MCA-3rd CBCS/NEP

Time : 3 Hours **Max. Marks : 60**

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

SECTION - A

1. (a) Define Software Engineering. Discuss various software myths. (4)

   (b) Discuss the Spiral Model of software development process in detail. (8)

2. (a) Which of the development process model will you follow for the following and why: (4)

   i) A simple data processing project.

   ii) A data entry system for the office who never used computer before.

   iii) A spreadsheet system with basic features and other features that use basic features.

   iv) A new missile tracking system.

   (b) Discuss, in detail, the Waterfall model of software development process. (8)

SECTION-B

3. (a) Consider a project to develop a full screen editor. The major components identified are (1) Screen edit (2) Command language interpreter (3) File input and output (4) Cursor movement and (5) Screen movement.

The sizes for these are estimated to be 4K, 2K, 1K, 2K and 3K, respectively, delivered source code lines. Use COCOMO model to determine:

   (i) Overall cost and schedule estimates (assume values for different cost drivers, with at least three of them being different from 1.0) Take values for 4 cost drivers as 1.15, 1.15, 0.86 and 1.07.

   (ii) Cost and Schedule estimates for different phases. (8)

   (b) How is Agile methodology different from traditional, software development? (4)

4. (a) Consider a project with the following parameters. (8)

   (i) External inputs: 10 with low Complexity, 15 with Average and 17 with high Complexity.

   (ii) External Outputs: 06 with low Complexity and 13 with high Complexity.

   (iii) External Inquiries: 03 with low Complexity, 04 with Average and 02 with high Complexity.

   (iv) Internal logical files: 02 with Average and 01 with high Complexity.

   (v) External Interface files: 09 with low Complexity.

   In addition to the above, system requires (i) Significant Data Communication, (ii) Performance is very critical, (iii) Designed code may be moderately reusable, (iv) System is not designed for multiple installations in different organizations. Other complexity adjustment factors are treated as average. Compute the function point for the same.

   (b) What is Scrum? What are its advantages? (4)

SECTION-C

5. (a) What do you understand by coupling and cohesion? What are various types of coupling and cohesion? (8)

   (b) What is the difference between sequence diagram and collaboration diagram? Illustrate with an example. (4)

6. (a) What is Object oriented design? Discuss, in detail, the various object oriented design concepts. (8)

   (b) A software has to be developed for automating the University result system. Draw use case diagram explaining all actors and flow of events. (4)

SECTION- D

7. (a) What do you mean by software metrices? Discuss, in detail, the various object oriented metrices. (8)

   (b) Write an explanatory note on Software Re-engineering. (4)

8. (a) Discuss various information flow metrics and use case-oriented metrics. (8)

   (b) Write an explanatory note on Configuration management. (4)

SECTION-E (Compulsory)

9. (a) What is the difference between functional and non-functional requirements? Give two examples of each. (2)

   (b) What is the role of documentation in software engineering? (2)

   (c) What is the role of management in software development? (2)

   (d) What is the use of UML in software engineering? (2)

   (e) Annual Change Traffic (ACT) for a software system is 15% per year. The development effort is 600 PMs. Compute an estimate for Annual Maintenance Effort (AME). If the life time of the project is 10 years; what is the total effort of the project? (2)

   (f) What is Regression testing? What is the need to perform regression testing? (2)